

Obstacle Avoidance for Small UAVs Using Deep Learning-Based Monocular Camera Depth Estimation

Original English title: Micro UAV Autonomous Navigation System With Deep Learning based Monocular Vision Depth Estimation

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Abstract. UAVs are increasingly used for search and inspection in cluttered environments. Collision-avoidance systems are therefore crucial for safe UAV navigation. This paper proposes a collision-avoidance navigation system using monocular depth estimation for lightweight UAVs. The system divides the depth map generated from a single RGB image by a deep-learning model into a grid and calculates the mean value of each cell to select the next flight position. The system is tested in a simulation environment, and its performance is demonstrated.

Keywords: Computer Vision, UAV System, Autonomous Navigation, Collision Avoidance

1. Introduction

Autonomous-flight technologies using drones play important roles in indoor inspection, search and rescue, and other applications. Small drones are particularly effective for exploring confined spaces because their small size provides rapid control response and a high degree of freedom. However, their limited payload and battery capacity restrict the use of heavy, energy-intensive sensors such as LiDAR.

Understanding the three-dimensional characteristics of the environment is very important for autonomous obstacle avoidance. Mapping technologies using LiDAR or stereo cameras [1] and deep-learning-based monocular camera depth-estimation technologies [2] have been developed for this purpose. Monocular cameras are lightweight and consume little power, so depth estimation using them is actively used in small robots. This technology applies deep learning to estimate relative distances between pixels in a single RGB image, providing depth information without additional sensors.

This paper proposes a system applicable to small drones that uses monocular camera depth estimation to avoid obstacles ahead.

2. Method

2.1 Monocular Camera Depth Estimation

A pretrained MiDaS deep-learning model [2] is used for monocular camera depth estimation. The model takes a single RGB image and estimates relative distances between pixels, as shown in the left image of Figure 1. The transformed image is represented as an 8-bit grayscale image, with each pixel having a value between 0 and 255. The source describes pixels at the closest distance as having a value of 0, representing white, and pixels at the farthest distance as having a value of 255, representing black.

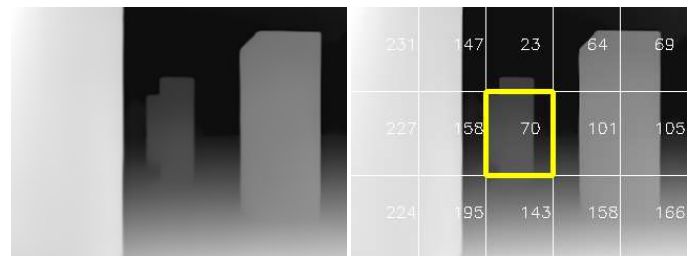


Figure 1. Selection of the target flight position using MiDaS.

2.2 Grid Division and Flight-Region Selection

The density of obstacles ahead is estimated to determine a safe flight path from the drone's current position. As shown in Figure 1, the depth image produced by the deep-learning model is divided into a grid with four cells horizontally and five cells vertically, and the mean depth value of each cell is calculated. The highest value indicates a high obstacle density in that region, while the lowest value indicates a low obstacle density.

The cell with the lowest depth value is selected to determine the flight path. Depending on the selected cell's position, the drone rotates its yaw by 10°, 5°, 0°, -5°, or -10°, from left to right. After completing the rotation at its current position, the drone advances 0.5 m in that direction.

2.3 Robot Operating System (ROS) Simulation

A ROS-based Gazebo simulation environment on Ubuntu 18.04 is built to validate the system. The drone model uses the open-source ROS package PX4-Autopilot. The MAVROS package controls MAVLink-compatible aircraft using ROS commands. The obstacle environment for the experiment is constructed by randomly placing rectangular pillars of dimensions 1 × 1 × 3 ahead of the drone, as shown in Figure 2.

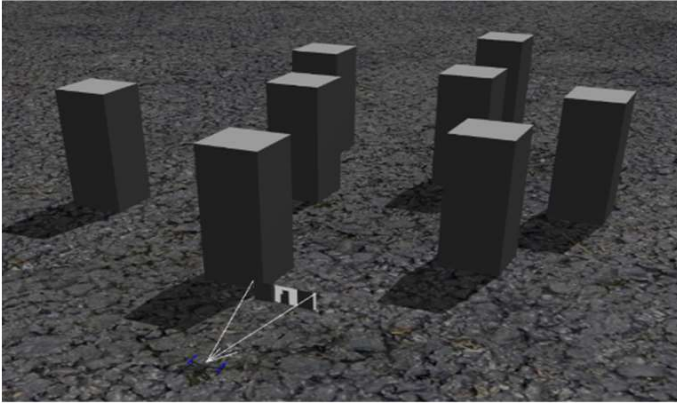


Figure 2. Configuration of the Gazebo simulation environment.

Figure 3 shows the overall system configuration. The depth map generated by MiDaS is sent to a module that calculates mean depth. The location of the minimum-value cell selected by this module is then sent to a module that calculates the drone's orientation and next position. The calculated orientation and position are transmitted through MAVROS to the drone in the simulation environment, causing it to follow the commands. The flight system proceeds through takeoff, flight toward the target position, and landing. During flight toward the target position, the target drone pose and flight position calculated in real time are received periodically and updated.

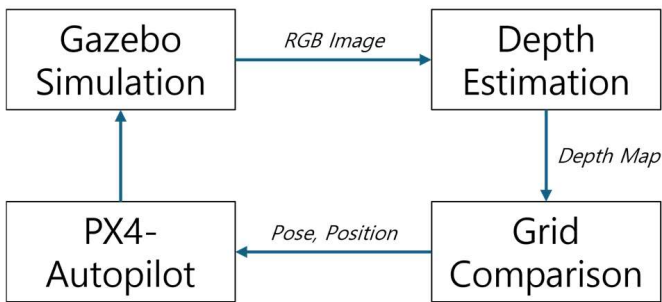


Figure 3. ROS system configuration.

3. Results

Experiments on monocular camera depth estimation and mean-depth processing confirmed that the deep-learning model could convert single RGB images into depth maps in real time at 41 FPS in the simulation environment.

Figure 4 shows the flight-experiment result. The blue line indicates the trajectory after the drone took off to a height of 1.5 m and completed a 10 m forward flight. Inspection of the flight path shows that the drone completed the flight without colliding with the obstacles ahead.

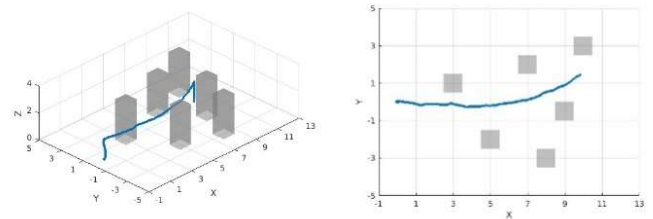


Figure 4. Flight path.

4. Conclusion

This paper developed an obstacle-avoidance flight technique for small drones using a monocular camera depth-estimation model. The drone selects its next target point from the mean depth values of the depth map generated by the deep-learning model. Simulation confirmed that the system selected a safe flight path and effectively avoided obstacles ahead. These results demonstrate the feasibility of obstacle-avoidance flight for small drones using a monocular camera.

Future work will further develop a path-generation algorithm that incorporates altitude control to create a system applicable to diverse environments. Performance will also be evaluated through field tests using an actual drone.

References

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Translation notes: (1) Section 2.1 pairs white with 0 and black with 255; conventional 8-bit grayscale uses black at 0 and white at 255. The source wording is preserved rather than silently corrected. (2) Section 2.2 states four horizontal by five vertical cells, while Figure 1 shows five columns by three rows. The five yaw angles are retained as written. (3) The source supplies no unit with the pillar dimensions $1 \times 1 \times 3$. (4) The acknowledgment has a typographical corruption ("2022sus"); its year is translated as 2022.